# Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | sahaj-bhadja | sahaj-bhadja | sahaj-bhadja

## Summary

Software engineer with 2+ years building full-stack applications, REST APIs, relational data services, and automated workflows using TypeScript, React, Python, Java, SQL, cloud platforms, containers, and AI-assisted development.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – June 2026*
- Architected multi-tenant Azure SQL and Blob Storage services for 500+ users, optimizing schemas, indexed queries, and access patterns to reduce application dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI workflows processing 5,000+ monthly jobs with explicit states, validation, reconciliation, monitoring, and failure isolation, achieving 99%+ reliability across recurring production application workflows.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*
- Developed React interfaces and Flask APIs handling 10,000+ daily requests, translating stakeholder requirements into intuitive upload, review, history, feedback, and reporting features for 150+ university instructors.
- Designed concurrent processing for 1,000+ files using multithreading, secure ingestion, AWS RDS, and fault-tolerant execution, debugging complex failure paths while maintaining stable performance during peak workloads.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*
- Engineered cross-platform services and 20+ automated PyTest and Mockito suites through Jenkins CI/CD, improving nightly release consistency across Windows, Linux, and macOS with detailed actionable failure reports.
- Optimized REST API and ETL workflows using object-oriented patterns, AWS Glue, S3, and SQL, improving response times by 30%, pipeline efficiency by 68%, and data-processing systems maintainability.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 – August 2023*
- Developed a modular MERN application with REST APIs, WebSockets, and eight services supporting 2M+ synchronized records while translating defined requirements into responsive customer and inventory workflows reliably.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving integration consistency, defect resolution, and production delivery reliability for engineering teams.

**LST Capitals** **India**
*Web Developer Intern* *November 2022 – January 2023*
- Built Flask APIs for a booking application supporting scheduling, payments, and status updates across 50,000+ records with synchronized Firebase-backed workflows, validation controls, and clear operational state tracking.
- Implemented webhook-based payment synchronization and root-cause troubleshooting, reducing manual handling by 70% while improving transaction visibility, error recovery, and responsiveness across distributed application components during production operations.

## Skills

**Programming & Frontend:** TypeScript, JavaScript, React, Angular, Next.js, Java, Python, C#, C++, HTML5, CSS3, responsive UI
**Backend & Architecture:** REST APIs, Flask, FastAPI, Node.js, microservices, WebSockets, multi-layer architecture, object-oriented design, data structures, algorithms
**Databases & Cloud:** SQL, PostgreSQL, SQL Server, Azure SQL, MySQL, Oracle, MongoDB, Firebase, AWS, Azure, GCP, Docker, Kubernetes
**Quality & Tools:** Git, GitHub, PyTest, Mockito, unit testing, integration testing, debugging, CI/CD, Jenkins, GitHub Actions, code reviews, documentation, Agile/Scrum, AI-assisted development

## Projects

**HMIS** Live | *TypeScript, MERN, REST APIs, Appwrite, Twilio*
- Built resident-style scheduling, record-access, and notification workflows supporting 100+ daily appointments, reducing administrative workload by 30% through responsive dashboards, automated reminders, and monitored API integrations across teams.

**Chat2SQL** Code | *Java, TypeScript, SQL, Docker, Maven, Node.js*
- Built an AI-assisted database workspace supporting 10+ systems, accelerating SQL authoring, schema exploration, validation, and data management through a unified TypeScript and Java interface for internal users.

**Linguistix** | *Python, AWS ECS/Fargate, Docker, CodePipeline, CI/CD*
- Developed a containerized data-processing API on AWS with automated CodePipeline releases, executable validation checks, and Docker builds, achieving 87% accuracy and production-grade reliability across repeatable integration environments.